Aaron Wang

aaronw5@uic.edu · (425) 623-8843 · aaronw5.github.io · Google Scholar · LinkedIn

Education

University of Illinois Chicago Chicago, IL — August 2022 – Present
Ph.D., Physics

University of Washington Seattle, WA — September 2018 – June 2022
B.S., Comprehensive Physics

Awards

DOE Office of Science Graduate Student Research (SCGSR) Award – funds thesis research at a DOE national lab — 2027

URA Visiting Scholar – Universities Research Association award for research at Fermilab — 2026

Chicagoland Computational Traineeship in High Energy Particle Physics (C2 the P2) – two years of full PhD funding for work at Fermilab — 2023 – 2025

IRIS-HEP Fellowship – scientific computing software for the particle physics community — 2021

Washington Research Foundation Nanofabrication Fellowship – quantum computing research — 2019

Research Experience

Amazon Web Services, Fundamental AI Research New York, NY — May 2026 – Present
Applied Scientist Intern

- Research whether LLM agents can follow a wall-clock time budget on long-horizon tasks and use that time effectively, including time spent in tool calls and code execution that the model cannot measure from its own output
- Built time-budgeted evaluations in Factorio and Kaggle machine-learning competitions (MLE-bench) that measure deadline adherence, how much of the budget agents use, and whether they change strategy when the budget changes
- Found that agents told only the deadline stop early, using about a quarter of their time; showing them a live clock raised task output 77% with no missed deadlines
- Developed a reinforcement learning method (GRPO policy optimization) that trains agents to follow the budget and choose a different strategy for short and long budgets (e.g. in Factorio, build less and harvest sooner when the budget is short)
- Introduced trajectory grafting, a training-data augmentation that splices together agent runs from different budgets; it nearly doubled task output under long budgets on unseen maps
- Encoded wall-clock time directly in the model: repurposed part of the rotary position embedding to rotate with elapsed seconds instead of token position, so the model can attend to time the way it attends to sequence order; raised on-time task completion from 6% to 50%
- Scaled training to multi-node GPU clusters with hundreds of parallel environments, including a 27B-parameter model trained across AWS regions
- First-author paper at the Resource-Aware Agentic AI Workshop @ NeurIPS 2026

Fermi National Accelerator Laboratory Batavia, IL — June 2023 – Present
Graduate Research Fellow

- Member and author of the CMS Collaboration at the CERN Large Hadron Collider
- Design efficient transformer architectures for real-time particle jet classification at the LHC trigger: a physics-inspired linear transformer that uses 80% less memory than full attention at matching accuracy, and a hierarchical patch-attention model with geometric message passing that reaches state-of-the-art accuracy among resource-constrained taggers (NeurIPS 2026)
- Built mixture-of-experts particle transformers and showed that routing each particle to a single expert beats a dense model of the same compute on a 188-class jet classification benchmark
- Explained what jet-tagging transformers learn: showed that their attention is nearly binary (each particle attends to one other) and traced the cause to the attention mechanism itself; rewrote the hidden layers of a deployed tagger as if-statements on physical jet quantities that reproduce the network's decisions on 94% of jets
- Deployed transformer models on FPGAs for the CMS Level-1 trigger: quantization-aware training and high-level synthesis to bit-exact RTL, reaching sub-100 ns inference within 10% of the device's logic budget
- Found that patch attention saves more FPGA logic than its FLOP count predicts (3$\times$ fewer lookup tables than full attention at 64 particles, against 1.5$\times$ by FLOPs) because it removes the activation-by-activation multiplies that need real multipliers; identified quantization, not architecture, as the accuracy bottleneck
- Built a hardware-aware neural architecture search that uses learned models of FPGA resource use and latency to find designs that fit the trigger budget (AutoML 2026)
- Trained models on 100-million-particle datasets with distributed training at Argonne National Laboratory; searched LHC data for rare dark matter interactions with an automated analysis framework

Amazon Web Services, Fundamental AI Research New York, NY — September 2025 – December 2025
Applied Scientist Intern

- Tested whether automatic prompt optimizers (GEPA, MIPROv2, APE), which rewrite an LLM's prompt from its own feedback, improve reasoning or only change output formatting and inject memorized facts
- Evaluated each optimized prompt over 10–30 repeated runs with statistical tests (paired t-test, Mann–Whitney U, McNemar's) across GPT-OSS, Claude Sonnet and Qwen3 on AIME math, BigBench Extra Hard and HotpotQA
- Found that prompt optimization gave no significant accuracy gain on 16 of 24 BigBench tasks; of the 8 that improved, 5 were explained by output formatting and 2 by memorized facts in the prompt, leaving 1 task with a reasoning gain
- Showed by ablation that the memorized facts were the source of the gains: removing them from an optimized prompt dropped accuracy back to that of a prompt with formatting changes only
- Found a second effect of memorized context: optimized prompts cut the share of answers that ran past the token limit (from 68% to 11% on one task) because the model no longer had to derive facts it could read from the prompt
- Tested the hypothesis that the optimizers failed on math because the training sets mix problem types: split the data into single-type subsets (algebra, counting, geometry) and found no improvement there either (0 of 272 GEPA-generated prompts), ruling out dataset diversity as the cause

University of Washington Particle Physics Group Seattle, WA June 2020 – July 2022
Research Assistant
- Applied state-of-the-art machine learning models to particle jet classification, reaching 99% accuracy
- Ranked input-feature importance for jet classification by pruning features on benchmark datasets
- Built autoencoder and GAN models to detect anomalous events at the LHC trigger
- Quantized recursive and recurrent neural networks for FPGA deployment with hls4ml

Pacific Northwest National Laboratory Richland, WA July 2021 – September 2021
Machine Learning Research Intern
- Built few-shot models in PyTorch to predict the solubility of materials
- Analyzed model errors by molecule type and related them to the underlying chemistry
- Cut the input features by 30% with no loss in accuracy, using Spearman correlation to rank feature importance

University of Washington Quantum Materials Lab Seattle, WA March 2019 – July 2022
Research Assistant
- Grew quantum material crystals by flux, vapor and Bridgman methods, including multiple dopings of MnBiTe and $PdTe_2$
- Measured resistivity versus temperature on prepared samples
- Designed and refined growth recipes for new materials with graduate students

IRIS-HEP New York, NY January 2021 – June 2022
Research Fellow
- Extended the Reproducible Open Benchmarks (ROB) platform, a system for comparing physics analyses on shared data, to run Jupyter notebooks; built with NYU computer science and physics faculty
- Built demos introducing ROB to the particle physics community

Washington Nanofabrication Facility Seattle, WA June 2019 – September 2019
Summer Research Fellow
- Fabricated cantilevers in a clean room, designing etch depths and masking layers and verifying them experimentally
- Used the cantilevers to measure fundamental frequencies and study structural transitions in quantum materials

PROJECTS

Distilling Jet Taggers into Physics Equations github.com/aaronw5/jet-tagger-distillation
- Rewrote the latent representations of two jet taggers, the FPGA-deployable real-time model JEDI-linear and the full-size Particle Transformer (ParT), as if-statements on jet physics quantities such as jet and subjet masses
- For JEDI-linear, the distilled formulas reproduce the network's decision on 94% of jets (64 particles per jet) with equal classification accuracy
- The same formulas explain 92% of each neuron's variance in the trained network versus 16% in an untrained copy

Interpretability Explorer for a Trigger-Level Jet Tagger github.com/aaronw5/jedi-linear-explorer
- Wrote every neuron of JEDI-linear, a quantized real-time jet tagger, as an exact formula of its particle inputs (p_T, $\Delta\eta$, $\Delta\phi$) and matched each to a jet physics quantity
- Built an interactive explorer with sparse autoencoders, transcoders and decision trees distilled from the network, evaluated on 50,000 held-out jets

Machine Learning for Cryogenic Transistor Modeling github.com/aaronw5/Cryo-Transistor-ML
- Trained neural-network surrogates of transistor current–voltage curves to extract compact-model (BSIM4) parameters for an open-source 130 nm CMOS process at cryogenic temperature (77 K), validating every fit in circuit simulation
- Lowered fitting error relative to the published parameters on all 18 test devices (combined relative RMS error 0.260 to 0.214)
- First-author paper at the Sim2Sci Workshop @ NeurIPS 2026

Selected Publications

Full list, including CMS Collaboration papers: Google Scholar

1. **Wang, A.**, Zhao, Z., Xia, A., Sun, C., Gandrakota, A., Ngadiuba, J., Cavanaugh, R., & Duarte, J. (2026). Patch Hierarchical Attention Transformer for Efficient Particle Jet Tagging. ***NeurIPS 2026***. arXiv:2605.21789.
2. **Wang, A.**, Madan, N., Sobal, V., Trager, M., Kleinman, M., Mansimov, E., Xia, W., & Soatto, S. (2026). On the Clock: Towards Punctual and Productive Time-Budgeted AI Agents. ***Resource-Aware Agentic AI Workshop @ NeurIPS 2026***. arXiv:2610.10833.
3. **Wang, A.**, et al. (2026). Surrogate Assisted Compact Model Extraction for Commercial CMOS Process Operating at Cryogenic Temperatures. ***Sim2Sci Workshop @ NeurIPS 2026***.
4. Weitz, J., Demler, D., Hawks, B., **Wang, A.**, Tran, N., & Duarte, J. (2026). SNAC-Pack 2.0: Scaled-Out Surrogate Neural Architecture Codesign. ***AutoML 2026*** (Oral). arXiv:2605.16138.
5. Pendiyala, K., Zia, H., Lee, T., Legge, T., De Leon, A. J., Zhao, Z., **Wang, A.**, Gandrakota, A., Ngadiuba, J., Cavanaugh, R., & Duarte, J. (2026). Conditional Capacity and Routing in Mixture-of-Experts Particle Transformers. *arXiv preprint* arXiv:2610.02701.
6. **Wang, A.**, Legge, T., Ortiz, J., Limouzi, V., Zhao, Z., Gandrakota, A., Khoda, E. E., Ngadiuba, J., Duarte, J., & Cavanaugh, R. (2025). Why Is Attention Sparse in Particle Transformer? ***Machine Learning and the Physical Sciences (ML4PS) Workshop @ NeurIPS 2025***. arXiv:2512.00210.
7. **Wang, A.**, Zhao, Z., Katel, S., Sahu, V. G., Khoda, E. E., Gandrakota, A., Ngadiuba, J., Cavanaugh, R., & Duarte, J. (2025). Spatially Aware Linear Transformer (SAL-T) for Particle Jet Tagging. *arXiv preprint* arXiv:2510.23641.
8. **Wang, A.**, Gandrakota, A., Ngadiuba, J., Sahu, V., Bhatnagar, P., Khoda, E. E., & Duarte, J. (2024). Interpreting Transformers for Jet Tagging. ***Machine Learning and the Physical Sciences (ML4PS) Workshop @ NeurIPS 2024***. arXiv:2412.03673.
9. Jiang, Q., Palmstrom, J. C., Singleton, J., et al. (2024). Revealing Fermi surface evolution and Berry curvature in an ideal type-II Weyl semimetal. ***Nature Communications*** 15, 2310. doi:10.1038/s41467-024-46633-w.
10. Khoda, E. E., Rankin, D., Teixeira de Lima, R., Harris, P., Hauck, S., Hsu, S.-C., Kagan, M., Loncar, V., Paikara, C., Rao, R., Summers, S., Vernieri, C., & **Wang, A.** (2023). Ultra-low latency recurrent neural network inference on FPGAs for physics applications with hls4ml. ***Machine Learning: Science and Technology*** 4, 025004. arXiv:2207.00559.

Talks and Teaching

Lecture and tutorial, "Transformers," Parts I and II, ML4FP School 2026, Georgia Tech, Atlanta, GA	June 2026
Talk, "Efficient and Interpretable Transformers for Jet Tagging," Fermilab AI Seminar Series, Batavia, IL	March 2026
Talk, "Spatially Aware Linear Transformer (SAL-T) for Efficient Particle Jet Identification," ML4Jets 2025, Caltech, Pasadena, CA	August 2025
Talk, "Fast RNN Inference on an FPGA," APS Division of Particles and Fields Meeting (DPF21), virtual	July 2021
Talk, "Implementation of Jupyter Notebooks into the Reproducible Open Benchmarks for Data Analysis Platform (ROB)," APS DPF21, virtual	July 2021
Talk, "Implementation of Jupyter Notebooks into the Reproducible Open Benchmarks for Data Analysis Platform (ROB)," ML4Jets 2021, virtual	July 2021
Talk, "Jupyter Notebook Compatibility with ROB," IRIS-HEP Fellow Presentations, virtual	June 2021

Posters

"Patch Hierarchical Attention Transformer for Efficient Particle Jet Tagging," NeurIPS 2026	December 2026
"On the Clock: Towards Punctual and Productive Time-Budgeted AI Agents," Resource-Aware Agentic AI Workshop @ NeurIPS 2026	December 2026
"Surrogate Assisted Compact Model Extraction for Commercial CMOS Process Operating at Cryogenic Temperatures," Sim2Sci Workshop @ NeurIPS 2026	December 2026
"Why Is Attention Sparse in Particle Transformer?," ML4PS Workshop @ NeurIPS 2025	December 2025
"Efficient Transformer Architectures for Jet Tagging," ACAT 2025, Hamburg, Germany	September 2025
"Efficient and Interpretable Transformers for Particle Physics," FastML 2025, ETH Zurich, Switzerland	September 2025
"Interpreting Transformers for Jet Tagging," ML4PS Workshop @ NeurIPS 2024	December 2024
"Interpreting and Accelerating Transformers for Jet Tagging" (with lightning talk), FastML 2024, Purdue University, West Lafayette, IN	October 2024

TECHNICAL SKILLS

LLM & ML Methods:	GRPO and RL post-training, RLHF, supervised fine-tuning, reward design, trajectory-level data augmentation, RoPE and architecture modification, hardware-aware neural architecture search, LLM agents and evaluation
ML Frameworks:	PyTorch, TensorFlow, Hugging Face, vLLM, SGLang, Megatron, verl, scikit-learn, NumPy, Pandas
Infrastructure:	AWS, multi-node GPU clusters, Docker, Kubernetes, distributed training, ROOT
Hardware ML:	Quantization-aware training (HGQ2, QKeras), hls4ml, da4ml, FPGA resource and latency estimation
Programming:	Python, Bash, Julia, MATLAB, Mathematica, R, Java, LaTeX
Mathematics:	Probability theory, statistics, linear algebra, differential equations, multivariable calculus

MENTORSHIP

Timothy Legge	Transformer interpretability; Ph.D. Physics, UC San Diego	Jan 2025 – Sep 2026
Victor Limouzi	Attention-free transformers; M.S. CS, Imperial College London	Feb 2025 – May 2026
Jacob Ortiz	Attention-free transformers; AI Builders Fellow, M.S. Data Science, UC Berkeley	Feb 2025 – May 2026
Trevin Lee	Mixture-of-experts particle transformers	Aug 2025 – Present
Haris Zia	Switch particle transformers	Aug 2025 – Present